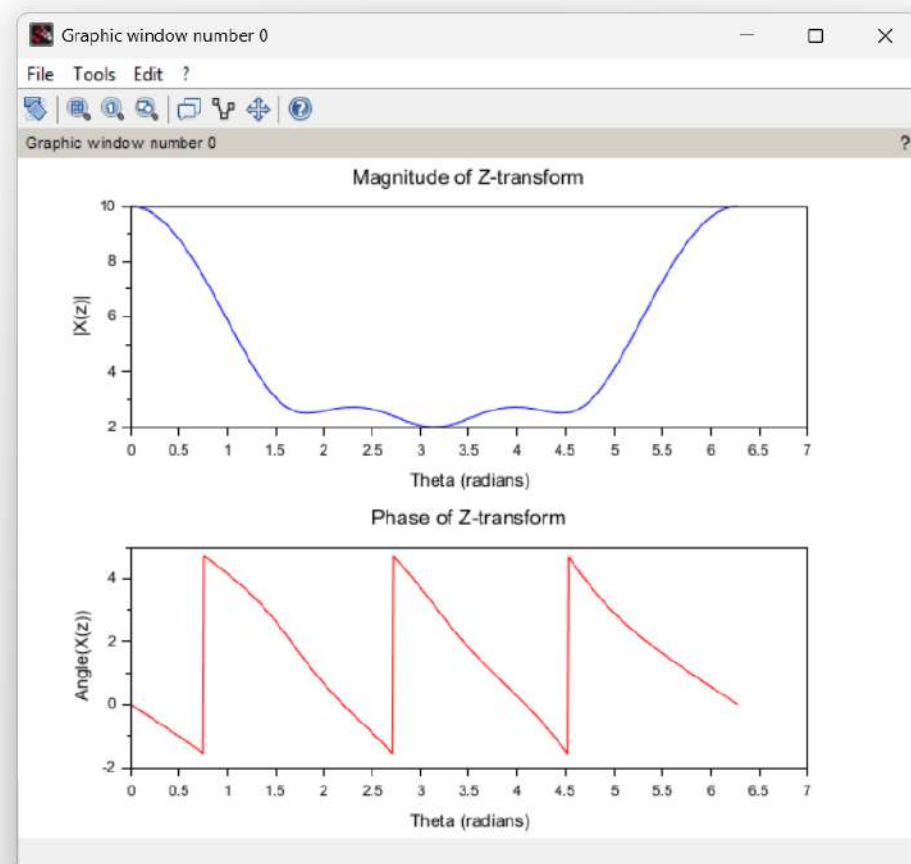


```

12 X_z = 0;
13 // Initialise Z-transform
14 for i = 1:length(n)
15 X_z = X_z + x_n(i) * z^(-n(i)); // Sum x[n] * z^(-n)
16 end
17 // Display the Z-transform expression
18 disp("Z-transform of x[n] = " + string(X_z));
19 // Define a range of z-values on the unit circle
20 theta = 0:0.01:2*pi; // Angle from 0 to 2*pi
21 z_vals = cos(theta) + %i * sin(theta); // Points on the unit circle
22 // Initialise arrays to store magnitude and phase
23 magnitude = zeros(1, length(theta));
24 phase = zeros(1, length(theta));
25 // Evaluate X(z) on the unit circle
26 for i = 1:length(z_vals)
27 z_val = z_vals(i);
28 X_z_val = 0;
29 for j = 1:length(n)
30 X_z_val = X_z_val + x_n(j) * z_val^(-n(j)); // Evaluate X(z) at z_vals(i)
31 end
32 magnitude(i) = abs(X_z_val); // Magnitude
33 // Calculate phase using atan
34 phase(i) = atan(imag(X_z_val) / real(X_z_val));
35 if real(X_z_val) < 0 then
36 phase(i) = phase(i) + %pi;
37 end
38 end
39 // Plot magnitude and phase of the Z-transform
40 clf; // Clear the current figure
41 // Plot magnitude
42 subplot(2,1,1);
43 plot(theta, magnitude, 'b');
44 xlabel('Theta (radians)');
45 ylabel('|X(z)|');
46 title('Magnitude of Z-transform');
47 // Plot phase
48 subplot(2,1,2);
49 plot(theta, phase, 'r');
50 xlabel('Theta (radians)');
51 ylabel('Angle(X(z))');
52 title('Phase of Z-transform');

```



```

1  clc;
2  clear;
3  //Time vector (0 to 5 sec, step 0.001 s)
4  t = 0:0.001:5;
5  dt = t(2) - t(1); //time step
6  //Define continuous-time signal x(t) = e^(-2t) * sin(3t)
7  x = exp(-2*t) .* sin(3*t);
8  //s-domain values
9  sigma = 0;
10 omega = -20:0.5:20;
11 //Damping factor
12 //Frequency range
13 //Initialize Laplace magnitude array
14 X_mag = zeros(1, length(omega));
15 //Numerical Laplace calculation (manual trapezoidal rule)
16 for k = 1:length(omega)
17     s = sigma + i*omega(k);
18     Xs = x .* exp(-s*t);
19     //trapezoidal integration
20     X_mag(k) = abs(sum(Xs(1:end-1) + Xs(2:end)) * dt ./ 2);
21 end
22 //Plot magnitude spectrum
23 clf;
24 plot(omega, X_mag, 'b');
25 xlabel("Frequency (rad/s)");
26 ylabel("|X(s)|");
27 title("Numerical Laplace Transform Magnitude");
28

```

